The Ice Decides
Solid–liquid equilibrium for freeze concentration and cryoprotectant design

Siddhant Chowdhary (25B0347)
Theme: Thermodynamics in Biological Systems

Abstract. Cryopreservation banks blood, cells and tissues. Better methods could one day store organs for transplant. Each new protectant recipe is still found by measurement. Slow freezing concentrates salt around and inside cells. Because ice is pure water, solid–liquid equilibrium fixes the water activity of every liquid touching ice. A recipe changes only how much water stays liquid and which solutes share it. To predict this without measuring every mixture, we model water activity with one constant per hydroxyl group, fitted on glycerol alone and tested blind. It predicted ethylene glycol, sucrose and maltose 3–17× more accurately than an ideal solution and beat UNIFAC for every solute except sorbitol. For glycerol + NaCl above 30 wt% solute it was 3–5× more accurate than two standard cryobiology models. It fails for sorbitol and simple alcohols. Sucrose at two temperatures traces this to the molecule and gives an interaction enthalpy of −6 to −8 kJ mol^{-1} per OH. The model shows that non-penetrating sorbitol lowers salt outside a cell but not inside. Whether it spares glycerol depends on the injury mechanism. Under three literature criteria it saves 0.76 g glycerol per gram, costs 2.1 g or allows an 11% reduction.

1. Introduction and background

Why it matters. Freezing is how blood, sperm, embryos and stem cells are banked. Better preservation of cells, tissues and whole organs would ease limits on transplantation, regenerative medicine and drug discovery [2]. Every protectant recipe is a compromise. Too little and salt or ice injures the cells. Too much and the protectant itself is toxic. Today each recipe is tuned by experiment, which is slow and costly.

Problem. In slow freezing, ice forms outside the cells first. The unfrozen solution concentrates and water leaves the cells. This solution injury, one arm of Mazur's two-factor account of freezing damage [1], acts between −15 and −60 °C [3]. Lovelock showed that glycerol protects red cells by diluting the salt that would otherwise accumulate [4, 5]. Mazur later found that the amount of liquid left matters as much as its salt content [6, 7]. In Pegg's words, "Cryoprotectants, simply by increasing the total concentration of all solutes in the system, reduce the amount of ice formed at any given temperature; but to be biologically acceptable they must be able to penetrate into the cells and have low toxicity" [8]. The first half of that sentence is thermodynamics. This work makes the second half quantitative too.

Thermodynamic principle. At equilibrium the chemical potential of water is the same in ice and in the liquid. Ice is pure, so $\mu_w^{\mathrm{ice}}(T) = \mu_w^{\mathrm{L}}(T) + RT\ln(x_w\gamma_w)$, which means $RT\ln(x_w\gamma_w) = -\Delta_{\mathrm{fus}}G(T)$. Since G is a state function, $\Delta_{\mathrm{fus}}G(T)$ can be evaluated along a heat–melt–cool path through T_m [9],

$$\Delta_{\mathrm{fus}}H(T) = \Delta_{\mathrm{fus}}H + \Delta C_p(T - T_m),$$
$$\Delta_{\mathrm{fus}}S(T) = \Delta_{\mathrm{fus}}H/T_m + \Delta C_p\ln(T/T_m),$$

and $\Delta_{\mathrm{fus}}G = \Delta_{\mathrm{fus}}H(T) - T\Delta_{\mathrm{fus}}S(T)$ gives

$$\ln(x_w\gamma_w) = -\frac{\Delta_{\mathrm{fus}}H}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right) - \frac{\Delta C_p}{R}\left(\frac{T_m}{T} - 1 + \ln\frac{T}{T_m}\right) \quad (1)$$

with $\Delta_{\mathrm{fus}}H = 6009.5\,\mathrm{J\,mol^{-1}}$, $\Delta C_p = 37.9\,\mathrm{J\,mol^{-1}K^{-1}}$ and $T_m = 273.15\,\mathrm{K}$. The right-hand side depends on T only, which has two consequences used throughout. (i) Once ice is present, the water activity a_w of the unfrozen liquid is fixed by temperature. Additives cannot change it. They change only how much water must freeze out to reach it. (ii) A cell at osmotic equilibrium has the same a_w inside, so its intracellular salt is set by which solutes can enter it. Everything else hinges on γ_w. For water–polyol mixtures $\gamma_w < 1$ because hydroxyl groups hydrogen-bond to water. A polyol therefore lowers the freezing point more than Raoult's law predicts.

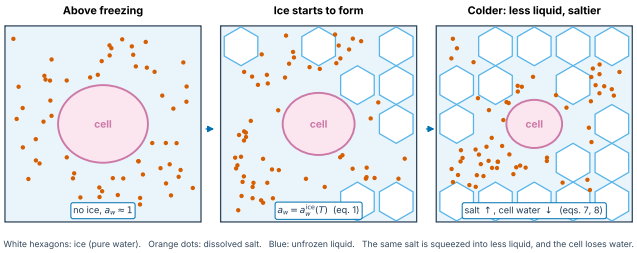


Figure 1: Freeze concentration. Ice is pure water, so the same solutes are squeezed into less liquid as T falls and water leaves the cell.

Existing approaches. Cryobiology uses ideal-dilute models, the osmotic virial equation [10, 11] and freezing-point summation [12]. The last two need coefficients fitted for every solute. Empirical phase-diagram equations exist for water–glycerol–NaCl [13], measured by thermal analysis [14]. Margules models have been fitted to water–glycerol SLE [15], so a glycerol fit alone is not new. Group-contribution methods (UNIFAC [16], Dortmund [17]) predict any molecule from parameters fitted mostly to vapour–liquid data. Polyol- and sugar-specific versions exist at 25 °C [18, 19]. A recent review suggests such models could screen cryoprotectant mixtures but does not apply them to freezing [20].

Why a predictive model. Every existing route to the freeze concentration of a new recipe starts with measurements. Either a phase diagram is measured for that mixture (Shepard *et al.* needed nine composition series by thermal analysis for one ternary system [14]) or freezing data are needed for every solute before the summation or virial models can be used. Screening candidate formulations this way multiplies experiments. We therefore

set three design criteria for the method itself. Cost means no new measurements to make a prediction, since the single parameter comes from published glycerol data. Time means a prediction is one root-finding step on a laptop and the whole study is rebuilt by one command. Feasibility means the inputs are textbook properties of ice, the number of OH groups and one constant, while validation needs only a few samples with calibrated temperature probes (Section 2.4).

Gap, hypothesis and contributions. Per-solute coefficients cannot be used for unmeasured molecules. General group parameters are not tuned to freezing. Our hypothesis is that one water–OH constant, fitted on glycerol only, predicts the freezing of other polyols and of polyol + salt mixtures without refitting. We test it blind against 13 solutes, two solutes at 25 °C and a ternary phase diagram, using cryobiology's own models as benchmarks. We then diagnose where and why it fails, including an enthalpy–entropy split of the interaction. Finally we use it to show how freeze concentration acts inside and outside a cell and what that implies for glycerol-sparing recipes. This combination was not found in the literature we searched.

2. Methodology

2.1 Thermodynamic framework

Every calculation is eq. (1) with a model for $\ln a_w$ of the liquid, solved numerically for the unknown (T or the amount of water left liquid).

Polyols. Only water–solute pair interactions are kept, as a one-constant Margules term for each solute proportional to its number of OH groups

$$\frac{G^{\mathrm{ex}}}{RT} = x_w \sum_i \alpha_i x_i, \qquad \alpha_i = \frac{n_{\mathrm{OH},i}\, a_{\mathrm{OH}}}{RT}. \tag{2}$$

Differentiating nG^{ex} gives the activity coefficients

$$\begin{aligned} \ln\gamma_w &= (1-x_w)\textstyle\sum_i \alpha_i x_i, \\ \ln\gamma_i &= \alpha_i x_w - x_w \textstyle\sum_j \alpha_j x_j, \end{aligned} \tag{3}$$

which satisfy Gibbs–Duhem, $x_w\,\mathrm{d}\ln\gamma_w + \sum_i x_i\,\mathrm{d}\ln\gamma_i = 0$ (numerical residual $\sim 10^{-14}$). Glycerol has 3 OH groups, ethylene glycol 2, sorbitol 6 and sucrose 8.

Salt. NaCl ($\nu = 2$) enters through an osmotic coefficient fitted to its CRC ice line [22] with RMSE 0.023 K to the eutectic. Its first term has the Debye–Hückel shape.

$$\ln a_w^{\mathrm{NaCl}} = -\nu m M_w \phi(m), \quad \phi = 1 - \frac{a\sqrt{m}}{1+\sqrt{m}} + bm + cm^2 \tag{4}$$

Mixtures. Contributions are added in $\ln a_w$ with no salt–polyol cross term

$$\begin{aligned} \ln a_w &= \ln x_w + (1-x_w)\textstyle\sum_i \alpha_i x_i \;-\; \nu\, m\, M_w\, \phi(m) \\ &= \ln a_w^{\mathrm{ice}}(T), \end{aligned} \tag{5}$$

where x runs over water and polyols and m is per kg water. This rule follows from the activity framework and was fixed before mixture data were seen. Summing freezing-point depressions [12] is reported only as a comparison. In the dilute limit eq. (5) reduces to $\Delta T = K_f \sum_j \nu_j m_j$ with $K_f = RT_m^2 M_w/\Delta_{\mathrm{fus}}H = 1.86\,\mathrm{K\,kg\,mol^{-1}}$, which checks the constants.

Fit. a_{OH} was fitted by least squares to 19 CRC glycerol freezing points between 2 and 40 wt% [22] by solving eqs. (1) and (3) for T. The result is $a_{\mathrm{OH}} = -2414.9\,\mathrm{J\,mol^{-1}}$ (bootstrap 95% interval -2461 to -2395, data scatter only). The interaction energy was assumed independent of T, so $\alpha_i \propto 1/T$. This was fixed before any test at another temperature.

Diagnostics. Away from ice, isopiestic data give the osmotic coefficient $\phi = -\ln a_w/(M_w \sum_j \nu_j m_j)$. Any measured a_w can be inverted for the constant a molecule implies, $a = RT\,(\ln a_w - \ln x_w)/(n_{\mathrm{OH}} x_s^2)$. Its temperature dependence can be split by Gibbs–Helmholtz

$$a(T) = h - Ts, \qquad h = \frac{\partial(a/T)}{\partial(1/T)}, \tag{6}$$

where h and s are the enthalpy and entropy of the water–OH interaction.

2.2 Blind tests and benchmarks

After the fit, predictions were committed to a git repository and never edited. Test data were entered only afterwards. Single solutes were CRC freezing points for ethylene glycol, mannitol, glucose, fructose, sucrose and four alcohols [22], then maltose and lactose in a later round [22], plus sorbitol by DSC at 53–64 wt% [23]. A cryoscope sorbitol set [24] was excluded because it implied $\phi \approx 1.14$ at 1 wt%, where ϕ must approach 1. At another temperature we used isopiestic data at 25 °C for sorbitol [25] and sucrose [26]. For mixtures we used the water–glycerol–NaCl liquidus [14]. It is published only as a figure, so it was digitised by hand (nine glycerol to NaCl ratios, 76 points) with the calibration checked against the NaCl–water eutectic. The pass criterion fixed in advance was RMS $\leq$ 0.5 K for total solute $\leq$ 30 wt%. Benchmarks were the ideal solution, UNIFAC and Dortmund [16, 17, 21] and for mixtures the osmotic virial equation and the Kleinhans–Mazur summation with published coefficients [11, 12].

2.3 Freeze concentration inside and outside a cell

A medium of water W_0 and solutes n_j (0.9 wt% NaCl plus additives) is cooled below its freezing point. Solutes stay in the liquid, so a fraction φ of the water freezes until

$$\ln a_w\big(\{n_j\},\,(1-\varphi)W_0\big) = \ln a_w^{\mathrm{ice}}(T), \tag{7}$$

with $\ln a_w$ from eq. (5) and the right-hand side from eq. (1). A cell at osmotic equilibrium (slow cooling) has the same a_w inside. Its own solutes n_c are represented as isotonic NaCl. Glycerol penetrates at the external molality $m_g(T)$ while sorbitol and NaCl are excluded. The cell water W_c then solves

$$\ln a_w\big(n_c/W_c,\; m_g(T)\big) = \ln a_w^{\mathrm{ice}}(T). \tag{8}$$

Recipes are designed at -15 °C [3] under three criteria from the literature. The first is NaCl outside the cell $\leq$ 4.41 wt%, the onset of red-cell haemolysis [5, 14]. The second is the same limit inside the cell. The third is unfrozen water $\geq$ 25%, above which most red cells survived regardless of salt [6]. For each criterion we report the sparing ratio $-\mathrm{d}w_g/\mathrm{d}w_s$, the glycerol saved per gram of sorbitol.

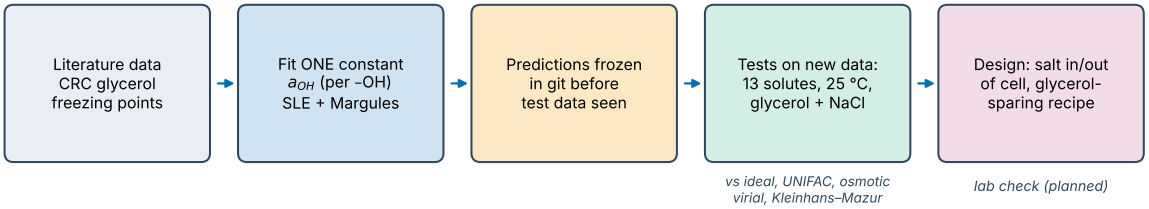


Figure 2: Workflow. One constant is fitted on glycerol, frozen in version control and tested against data it has never seen. Only then is it used for design. Every figure and table is rebuilt by one command (`run_all.py`).

Table 1: Proposed validation. Committed predictions of the freezing point (°C).

Sample (wt%)	Per-OH	Ideal gly.	K–M	Osm. virial
NaCl 5 (standard)		CRC −3.05		
Glycerol 30 (standard)	−9.69	−7.87	–	–
Glycerol 20 + NaCl 5	−9.42	−8.68	−9.69	−9.92
Glycerol 25 + NaCl 5	−11.71	−10.45	−12.11	−12.42

2.4 Proposed experimental validation

Laboratory access was not available before submission, so the experiment is proposed with its predictions already committed. The available reagents (NaCl and glycerol) are enough. Freezing points would come from cooling curves with calibrated probes in an ice–NaCl bath (to about −18 °C, above the −21.1 °C eutectic), taking the post-recalescence maximum corrected for supercooling. This follows the thermal-analysis approach of [14] and the re-measurement strategy of [30]. Samples and committed predictions are in Table 1. An uncertainty budget (probe calibration, nonlinearity, weighing, purity, supercooling and repeatability) gives ±0.25 K at 95%. That is enough to resolve our model from the ideal one (0.7–1.3 K apart) but not the three mixing rules (0.2–0.7 K).

3. Results and discussion

3.1 Where one constant per OH holds

For ethylene glycol, glycerol, sucrose and maltose a constant fitted on glycerol predicts freezing without refitting and does as well as osmotic virial coefficients fitted to each molecule (Table 2, Fig. 3a). Lactose dissolves only to 8 wt%, which is too dilute to discriminate. Fig. 4 draws the blind predictions over Shepard's original phase diagram. On this liquidus the model missed the pre-set 0.5 K target at low concentration (0.69 K). There the digitised figure is itself uncertain by 0.34 K (its pure-glycerol and pure-NaCl edges against CRC) and all three mixture models agree within 0.1 K. Above 30 wt% solute, which are the conditions cells meet during freezing, adding $\ln a_w$ is about 3× more accurate than the cryobiology models (Fig. 3b). Isopiestic sorbitol + NaCl data [25] show why. The neglected salt–polyol cross term (salting-in) is 0.6–5% of the osmolality, or 0.05–0.13 K at recipe conditions. A virial-type cross coefficient would remove this, but it needs ternary data for every salt–polyol pair, which is the per-mixture measurement this method avoids. With the error under a fifth of the model's own, the additive rule is the better choice for screening. A cross term is worth adding only to refine a final recipe.

3.2 Where it fails and what the failure teaches

The model overstates sorbitol's non-ideality by 3.5–5.5 K when concentrated and four- to five-fold in $\phi - 1$ at 25 °C (Fig. 7). Two causes were possible and sucrose separates them. On the freezing line sucrose implies $a \approx -2200\,\mathrm{J\,mol^{-1}}$ per OH, close to glycerol's. At 25 °C it implies about −1800. Sorbitol at the same 25 °C implies only −400 to −750. The failure is mainly the molecule. Post hoc (after the blind tests), letting each molecule have its own constant shows a pattern (Fig. 3c). Short acyclic polyols sit at $\approx -2.4\,\mathrm{kJ\,mol^{-1}}$ while six-OH chains (mannitol, sorbitol) fall to −0.9 to −1.3. Adjacent hydroxyls on a flexible chain can bond to each other rather than to water, so counting them independently overstates their pull on water. A proximity correction of this kind is used for sugars [19]. The simplest version subtracts a term per pair of OH groups on neighbouring carbons, $A_i = n_{\mathrm{OH},i} a_{\mathrm{OH}} - n_{\mathrm{pair},i}\,\delta$. The post hoc constants already show it cannot be adjacency alone. Ethylene glycol and glycerol (0.50 and 0.67 neighbouring pairs per OH) share $a \approx -2.45\,\mathrm{kJ\,mol^{-1}}$ while sorbitol (0.83) needs −1.3, so chain length or conformation must enter. A δ fitted to sorbitol would remove its deviation by construction, so we do not claim it. The test is xylitol, maltitol and lactitol [27], whose blind predictions are committed.

Gibbs–Helmholtz split (post hoc, Fig. 6). Applying eq. (6) to sucrose at both temperatures over matched molality windows gives $h = -6.3$ to $-7.6\,\mathrm{kJ\,mol^{-1}}$ and $s = -15$ to $-20\,\mathrm{J\,mol^{-1}K^{-1}}$ per OH. The water–OH interaction is strongly exothermic, consistent with hydrogen bonding. It is partly offset by an entropy loss as water is ordered around the solute. Because $|Ts|$ grows with T the net attraction weakens on warming. This is why the constant-energy assumption ($\alpha \propto 1/T$) overstates non-ideality at 25 °C and why our freezing-line fit should not be used for room-temperature water activity.

Ours versus UNIFAC. UNIFAC's many group parameters were fitted mostly to vapour–liquid data at and above room temperature. For water–glycerol it gives the wrong sign ($\gamma_w > 1$) and misses the depression by up to about 6 K at 40 wt%. One group and one constant fitted on freezing data trade breadth for accuracy within one class of molecules.

3.3 What freeze concentration does inside the cell

Without a cryoprotectant the unfrozen medium of saline holds 18.8 wt% NaCl at −15 °C and only 3.9% of the water is still liquid (Fig. 5a,b). Glycerol 6.5 wt% or sorbitol 9 wt% each bring the salt outside the cell down to the 4.41 wt% haemolysis limit. Inside the cell they differ completely (Fig. 5c). Glycerol enters the cell, shares the osmotic load and keeps intracellular salt at 4.4 wt%. Sorbitol stays outside, so the cell's interior still contains only its own salts. These must reach the a_w set by the ice, which gives 18.8 wt%, the same as in plain saline. This is consequence (ii) of eq. (1) and it turns Pegg's "must penetrate" [8] into a number.

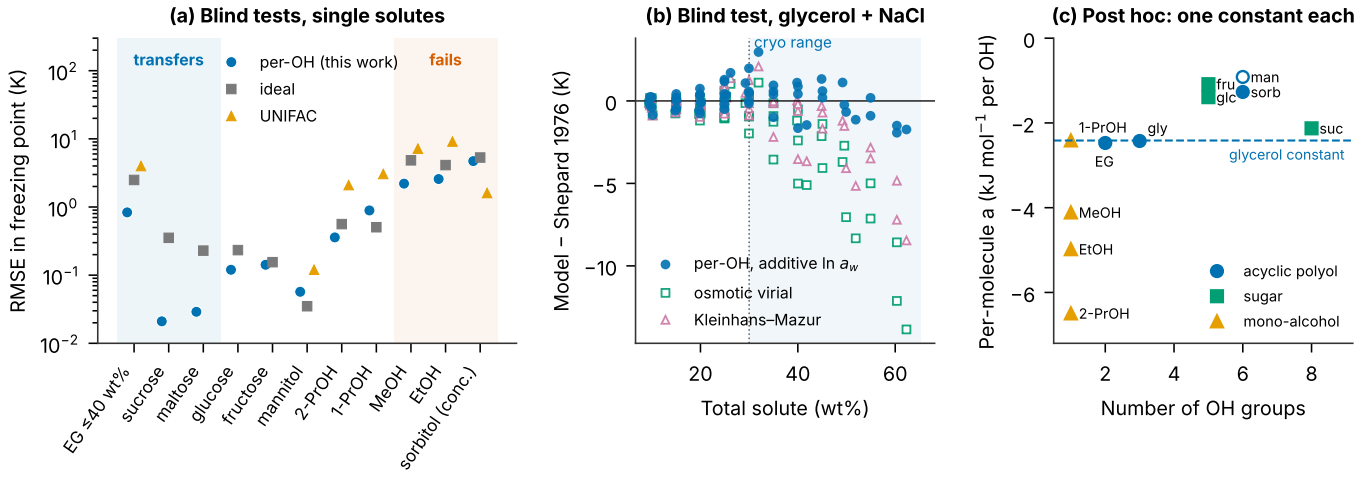


Figure 3: (a) Freezing-point error on single solutes not used in the fit (log scale). The blue band marks molecules where the constant transfers and the red band where it fails. (b) Error on the water–glycerol–NaCl liquidus [14]. All three models predict from single-solute data. (c) Post hoc constant each molecule would need on its own (fits ≤ 20 wt%). Open symbols mark depressions below 3 K, where the constant is weakly determined. The dashed line is the glycerol constant used in the blind tests.

Table 2: Blind tests (error = model − data, RMSE in K unless stated). a_{OH} was never refitted.

Test	Data	Per-OH	Ideal	UNIFAC	Other	Verdict
Glycerol (fit)	CRC [22]	0.02	1.31	2.1	0.27[a]	fit
Ethylene glycol, 2–40 wt%	CRC	**0.83**	2.49	3.99	0.52[a]	transfers
Sucrose, to 40 wt%	CRC	**0.02**	0.35	–	0.02[a]	transfers
Maltose, to 40 wt% (later round)	CRC	**0.03**	0.23	–	–	transfers
Glucose / fructose	CRC	0.12/0.14	0.23/0.15	–	0.01/–[a]	weak test
MeOH / EtOH	CRC	2.20/2.57	4.84/4.12	7.2/9.1	1.29/0.91[a]	fails
Sorbitol, 53–64 wt%	DSC [23]	4.7	5.3	**1.6**	–	fails
Sorbitol 25 °C (RMS in ϕ)	isopiestic [25]	0.163	**0.060**	0.113	–	fails
Glycerol + NaCl, ≤30 wt% solute	digitised [14]	0.69	0.89[b]	–	0.69[a]/0.59[c]	misses 0.5 K target
Glycerol + NaCl, >30 wt% solute	digitised [14]	**1.23**	4.64[b]	–	5.75[a]/3.39[c]	best

[a]Osmotic virial [11], fitted per solute (in-sample for single solutes). [b]Ideal glycerol + fitted NaCl. [c]Kleinhans–Mazur summation [12].

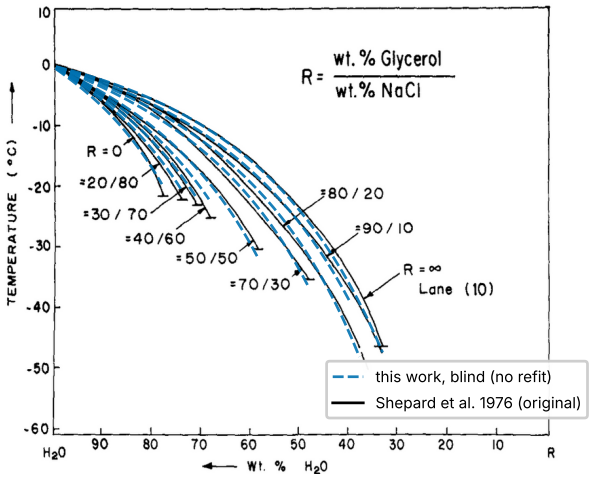


Figure 4: Water–glycerol–NaCl liquidus isopleths reproduced from Shepard *et al.* [14] (Fig. 3, black) with this work's blind predictions drawn on top (blue dashed). Only the glycerol-only curve ($R = \infty$) corresponds to the data used in the fit.

Design depends on the injury mechanism (Fig. 5d). If injury comes from salt outside the cell [5], sorbitol spares 0.76 g of glycerol per gram. This holds across three activity models (0.68–0.91 g/g, ±25% salt limit, −10 to −20 °C). If it comes from salt inside, sorbitol costs 2.1 g of glycerol per gram, because it keeps more water liquid outside and so dilutes the glycerol that must enter the cell. If it comes from too little liquid [6], sorbitol helps (0.66 g/g). Requiring all three at once gives an interior optimum of 2.1 wt% sorbitol with 10.9 wt% glycerol, 11% less glycerol than glycerol alone (12.2 wt%). Mazur's experiments suggest that which criterion dominates depends on cell concentration and on how much water freezes [6, 7]. The model makes that trade-off explicit. For red cells cooled slowly in saline to −15 °C, sorbitol is worth adding only up to about 2 wt%, beyond which the cell interior becomes the weak point.

Scope. The results are equilibrium predictions. They assume slow cooling and model the cell's solutes as isotonic NaCl. Cooling-rate kinetics, intracellular ice and toxicity are not included. The model predicts the conditions a cell experiences, not whether it survives.

4. Conclusion and next steps

A single hydroxyl constant fitted on one molecule predicts the freezing of small polyols, sucrose, maltose and glycerol + salt mixtures without refitting. It matches per-molecule cryobiology models when dilute and beats them when concentrated. It fails for six-OH chains and simple alcohols. These failures are diagnostic. The post hoc constants point to neighbouring-OH interactions and the sucrose data split the interaction into a hydrogen-bond enthalpy and an ordering entropy. Because ice fixes water activity everywhere, a non-penetrating additive cannot protect a cell's interior. Whether it spares glycerol depends on which injury mechanism dominates. The model puts a number on each case. In practice this is a cheap and fast screen. Recipes are ranked from textbook data and one constant. The laboratory is needed only to confirm the chosen few and for molecules such as six-OH chains where the screen fails.

Next steps. (1) Run the proposed validation (Section 2.4). (2) Add a hydroxyl-proximity term and test

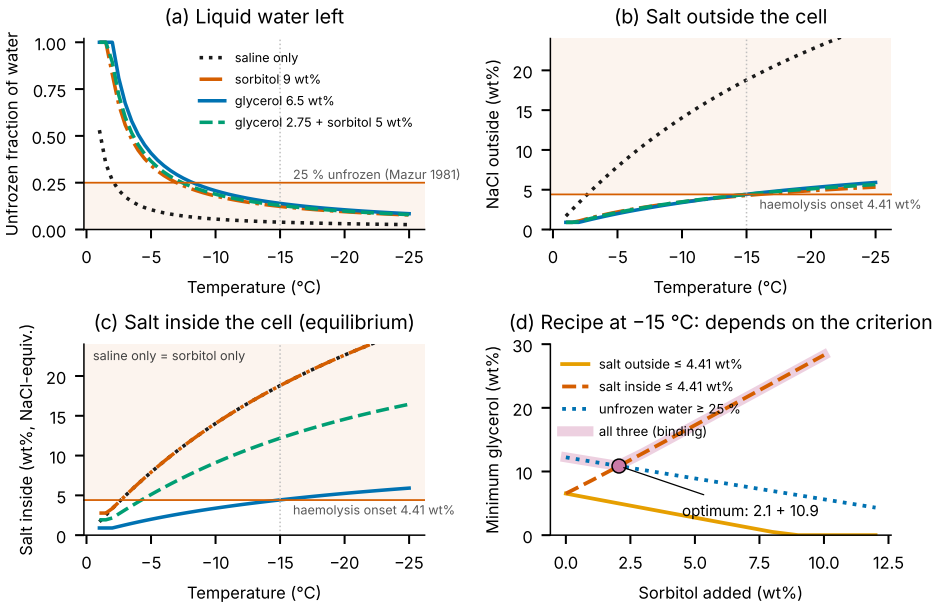


Figure 5: Freeze concentration in 0.9 wt% saline with the per-OH model. (a) Unfrozen fraction of the water. (b) NaCl in the unfrozen medium. (c) Salt inside a cell at osmotic equilibrium. Glycerol penetrates and sorbitol does not, so the saline-only and sorbitol-only curves coincide. (d) Minimum glycerol at −15 °C against sorbitol added under three literature criteria. The purple band shows all three at once with its optimum.

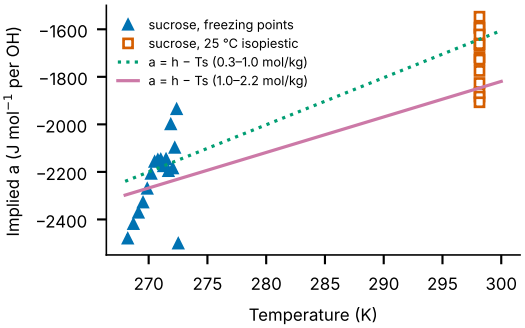


Figure 6: Post hoc per-OH constant implied by sucrose data near 0 °C and at 25 °C, with $a = h - Ts$ fitted on two matched molality windows.

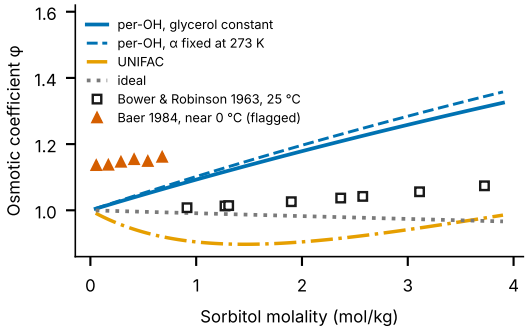


Figure 7: Sorbitol at 25 °C [25]. With glycerol's constant the per-OH model overstates non-ideality. An older freezing-point set [24] is implausible at infinite dilution, where ϕ must tend to 1.

it on xylitol, maltitol and lactitol. (3) Check h against calorimetric mixing enthalpies. (4) Couple the framework to membrane transport for finite cooling rates. In the two-parameter model [29] cell water and glycerol follow $\mathrm{d}V_w/\mathrm{d}t = L_p A\,(RT/\bar{V}_w)\ln(a_w^{\mathrm{out}}/a_w^{\mathrm{in}})$ and $\mathrm{d}n_g/\mathrm{d}t = P_s A\,(c_g^{\mathrm{out}} - c_g^{\mathrm{in}})$, with a_w^{out} from eq. (1) at each instant of cooling and a_w^{in} from eq. (5). This replaces the dilute osmolality of the classical treatment [28] with the non-ideal activity. That matters most when the cell is most dehydrated.

Code and data. Models, data with sources, committed predictions and all analyses are at `https://github.com/Siddhant-Chowdhary/Thermodynamics-Project-`. The README maps every figure and table to its script. `python run_all.py` rebuilds them all and `run_in_colab.ipynb` runs the same pipeline in a browser. An interactive version (`app.py`, online at `https://the-ice-decides.streamlit.app`) lets a reader pick a recipe and a temperature and see the unfrozen water and the salt inside and outside the cell.

References

[1] P. Mazur, *Am. J. Physiol.* 247 (1984) C125.
[2] S. Giwa et al., *Nat. Biotechnol.* 35 (2017) 530.
[3] D. Gao, J. K. Critser, *ILAR J.* 41 (2000) 187.
[4] J. E. Lovelock, *Biochim. Biophys. Acta* 10 (1953) 414.
[5] J. E. Lovelock, *Biochim. Biophys. Acta* 11 (1953) 28.
[6] P. Mazur, W. F. Rall, N. Rigopoulos, *Biophys. J.* 36 (1981) 653.
[7] P. Mazur, K. W. Cole, *Cryobiology* 26 (1989) 1.
[8] D. E. Pegg, *Methods Mol. Biol.* 368 (2007) 39.
[9] S. I. Sandler, *Chemical, Biochemical, and Engineering Thermodynamics*, 4th ed., Wiley, 2006.
[10] J. A. W. Elliott et al., *J. Phys. Chem. B* 111 (2007) 1775.
[11] M. W. Zielinski et al., *Cryobiology* 69 (2014) 305.
[12] F. W. Kleinhans, P. Mazur, *Cryobiology* 54 (2007) 212.
[13] G. M. Fahy, *Biophys. J.* 32 (1980) 837.
[14] M. L. Shepard, C. S. Goldston, F. H. Cocks, *Cryobiology* 13 (1976) 9.
[15] L. Zargarzadeh, J. A. W. Elliott, *J. Phys. Chem. B* 123 (2019) 1099.
[16] Aa. Fredenslund, R. L. Jones, J. M. Prausnitz, *AIChE J.* 21 (1975) 1086.
[17] U. Weidlich, J. Gmehling, *Ind. Eng. Chem. Res.* 26 (1987) 1372.
[18] C. Marcolli, Th. Peter, *Atmos. Chem. Phys.* 5 (2005) 1545.
[19] A. M. Peres, E. A. Macedo, *Fluid Phase Equilib.* 139 (1997) 47.
[20] M. J. Powell-Palm, A. N. Consiglio, arXiv:2601.12573 (2026).
[21] C. Bell et al., *thermo* (Python package) v0.6.1, `https://github.com/CalebBell/thermo`.
[22] D. R. Lide (ed.), *CRC Handbook of Chemistry and Physics*, Internet Version 2005, pp. 8-58 to 8-84.
[23] M. Siniti, S. Jabrane, J. M. Létoffé, *Thermochim. Acta* 325 (1999) 171.
[24] R. J. Baer, K. A. Baldwin, *J. Dairy Sci.* 67 (1984) 2860.
[25] V. E. Bower, R. A. Robinson, *J. Phys. Chem.* 67 (1963) 1540.
[26] R. A. Robinson, R. H. Stokes, *Trans. Faraday Soc.* 45 (1949) 612.
[27] T. Uraji et al., *Food Sci. Technol. Int.* 2 (1996) 38.
[28] P. Mazur, *J. Gen. Physiol.* 47 (1963) 347.
[29] F. W. Kleinhans, *Cryobiology* 37 (1998) 271.
[30] J. D. Benson et al., *Cryobiology* 61 (2010) 352.